

# Final Project: Contact Book CLI Application

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## Project Description

Create a Contact Book application that runs entirely in the terminal (command-line interface).

Users should be able to add, search, update, delete, and view contact information. The app must store contacts in a file so data persists across runs.

This simulates a basic database and CRUD (Create, Read, Update, Delete) system.

## Features

The application must allow users to:

- Add a contact
- View all contacts
- Search for a contact
- Edit a contact
- Delete a contact
- Save and exit

## Terminal User Interface

When the user runs the program, they should see a menu like this:

=====

```
=====
Contact Book
=====
1. Add Contact
2. View All Contacts
3. Search Contact by Name
4. Edit Contact
5. Delete Contact
6. Save and Exit

Choose an option (1-6):
```

## Inputs and Outputs

### Add Contact

Prompt user:

Enter full name: John Doe

Enter phone number: +123456789

Enter email address: john.doe@email.com

Output:

Contact "John Doe" added successfully!

### View All Contacts

If there are saved contacts, display like this:

```
==== All Contacts ====
1. John Doe | +123456789 | john.doe@email.com
2. Alice Smith | +987654321 | alice.smith@email.com
=====
Total: 2 contacts
```

If no contacts are found:

No contacts found.

### Search Contact by Name

Prompt:

Enter name to search: alice

Output:

Found 1 result:

```
1. Alice Smith | +987654321 | alice.smith@email.com
```

If no match:

No contact found with name containing "alice".

### Edit Contact

Prompt:

Enter name of contact to edit: John Doe

If found, display:

Current contact info:

Name: John Doe

Phone: +123456789

Email: john.doe@email.com

Enter new name (leave blank to keep current): Jonathan Doe

Enter new phone (leave blank to keep current):

Enter new email (leave blank to keep current):

```
jon.doe@newmail.com  
Contact updated!
```

## Delete Contact

Prompt:

```
Enter name of contact to delete: Alice Smith
```

Output:

```
Are you sure you want to delete "Alice Smith"? (yes/no): yes  
Contact deleted!
```

If not found:

```
Contact not found.
```

## Save and Exit

Output:

```
Saving contacts to file...  
Data saved. Goodbye!
```

## Data Format

Use a JSON file (contacts.json) to store the data persistently.

Structure:

```
[  
  {  
    "name": "John Doe",  
    "phone": "+123456789",  
    "email": "john.doe@email.com"  
  },  
  {  
    "name": "Alice Smith",  
    "phone": "+987654321",  
    "email": "alice.smith@email.com"  
  }  
]
```

## Technical Requirements

- Use a while loop to keep the menu running until exit
- Use a list of dictionaries or a list of Contact objects
- Use json for reading/writing contact data
- Use functions to modularize your code: add\_contact(), view\_contacts(), search\_contact(), edit\_contact(), delete\_contact(), save\_contacts(), load\_contacts()

## Error Handling

- Ensure input is not empty
- Check for duplicate names when adding
- Handle FileNotFoundError when loading contacts
- Validate email format and phone number (basic check)

## Optional Enhancements

- Sort contacts alphabetically when viewing
- Validate phone numbers
- Use a Contact class (OOP)
- Export contacts to .csv
- Color-code status messages using colorama

## Example Session

```
=====
Contact Book
=====
1. Add Contact
2. View All Contacts
3. Search Contact by Name
4. Edit Contact
5. Delete Contact
6. Save and Exit
Choose an option (1-6): 1
Enter full name: Alice Smith
Enter phone number: +987654321
Enter email address: alice@email.com
Contact "Alice Smith" added successfully!
Choose an option (1-6): 2
==== All Contacts ====
1. Alice Smith | +987654321 | alice@email.com
=====
Total: 1 contacts
Choose an option (1-6): 6
Saving contacts to file...
Data saved. Goodbye!
```

## What to Submit

A zip folder with your name including:

- contact\_book.py

- contacts.json (your saved data file)
- A short README.txt with: How to run your program, What features are implemented, Any extra functionality added
- The Github link if you have upload the file there

*Ideally, the project should be hosted on GitHub, and the GitHub repository link should be provided along with the submitted files. However, submission must always include the project files themselves.*